

Yanbao Li

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Education

Stony Brook University B.S. studies in Information Systems	Stony Brook, New York Jul 2026 - Present
Anhui University B.S. studies in Digital Media Technology (Transferred) Relevant Experience: Co-Lead, Mathematical Modeling Club; Teaching Assistant, Probability Theory	Hefei, China Aug 2024 - Jun 2026

Technical Skills

Languages: Python, Java, C#, TypeScript, Dart, SQL, MATLAB
Tools: Spring Boot, Fastify, Flutter, PyTorch, Git, MySQL, SQLite, Supabase, FFmpeg, yt-dlp
ML & Modeling: Data organization and cleaning, feature processing, model fitting, parameter tuning, experiment design, result analysis

Projects & Leadership Experience

VideoHarvester, Personal Project • Built a C# / Windows Forms workflow with yt-dlp and FFmpeg for authorized video-page links and playlist jobs. • Wrapped yt-dlp and FFmpeg in asynchronous process execution and output parsing, with preflight checks for title, quality, size, disk space, and save location. • Built batch-queue progress tracking, platform-ID deduplication, persistent history, unfinished-task recovery, and copyable diagnostic logs.	2026
AdMind, Personal Project • Built a TypeScript / Fastify long-video ad-decision prototype; normalized TwelveLabs time-coded analysis into cacheable, traceable risk evidence validated with Zod contracts. • Designed a deterministic decision engine that combines video semantics, ethical boundaries, contract constraints, and user-experience rules into executable, auditable plans. • Implemented a pause-state machine and browser-side MediaPipe four-corner occlusion scoring to choose safer placement or defer delivery when no safe area exists.	2026
Sona, Personal Project • Developed an offline-first Flutter / Dart music and MV player for Windows and Android; a single playback coordinator owns the authoritative track, queue, mode, and progress state. • Designed a local import pipeline that identifies media type, reads tags/duration/artwork, cleans filenames, and writes structured records to a SQLite catalog. • Implemented content-hash-based sync for tracks, playlists, favorites, and listening events, with diff preview, cross-device state merging, request generations, cancellation, and fallback safeguards.	2026
Multimodal Medical Image Fusion Research, Research Project • Organized, cleaned, normalized, and integrated multimodal medical-image data to support model training, evaluation, and repeatable experiments. • Supported data fitting, parameter tuning, and experiment runs on existing PyTorch models while consolidating outcomes across configurations for comparison.	Nov 2024 - Aug 2025
CUMCM Drone Deployment Optimization, Competition Project • Led the team's programming and modeling work; built a unified 3D kinematic model for drones, smoke rounds, smoke clouds, and missiles with line-of-sight-to-sphere occlusion checks. • Parameterized heading, speed, release time, and fuse delay; used time-discretized simulation to calculate effective coverage and generate scenario visualizations and animations. • Combined simulated annealing, Bayesian hyperparameter tuning, genetic algorithms, island-style parallel search, and NSGA-II for coverage, resource, and energy tradeoffs.	2025
MCM/ICM Cybersecurity Modeling, Competition Project • Led the team's primary programming and modeling work, integrating Global Cybersecurity Index, cybercrime, GDP, population, education, and internet-usage data. • Reconciled inconsistent time granularity and country names, then cleaned, merged, and filtered valid records to form a unified dataset for fitting and comparison. • Built Python/Folium/Choropleth cybersecurity maps and crime trends; supported clustering and Random Forest model fitting.	Feb 2025

Awards

CUMCM Mathematical Modeling Competition - Third Prize	2025
ICM Mathematical Contest in Modeling - Successful Participant	2025
Innovation & Entrepreneurship Competition - Excellence Award	2025